Sahaj Bhadja

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Summary

Forward Deployed Engineer with 2+ years shipping production **LLM** systems in **Python**, including **RAG** over 50K+ documents, multi-model pipelines, and evaluation, with Git, CI/CD, testing, and client-facing delivery to enterprise stakeholders.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Advanced NLP, Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Software Engineering, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Forward Deployed Engineer *Los Angeles, CA*

- Shipped a production multi-tenant **AI education platform** on Azure SQL and Blob Storage with row-level security for 500+ users, cutting dashboard latency from 1.2s to 350ms through query tuning.
- Rebuilt async **LLM orchestration** with FastAPI workers, explicit job states, and failure isolation, processing 1K+ AI jobs at 3x throughput while CI/CD-backed integrations held 99%+ production reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Turned a manual grading process into a production **LLM pipeline** chaining rubric generation, OCR, scoring, and feedback across GPT-4o, Claude, and Qwen-VL for 150+ instructors at 90%+ consistency.
- Built provider fallback routing and per-call token and **cost tracking**, scaling to 10K+ daily requests and 1K+ file batches with GitHub Actions CI/CD while cutting feedback turnaround 60%.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Trained a GPT-based Decision Transformer with offline **reinforcement learning** on 100K+ trajectory logs in PyTorch, tracking experiments in Weights & Biases and improving long-horizon planning 18% in simulation.
- **Evaluated** a spatial-temporal attention encoder against three baselines in closed-loop simulation, using the results to select the HD-map-aware design that improved closed-loop planning stability 15% and produced fewer off-road trajectories.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Automated cross-platform workflows that cut manual data entry 35%, working with Microsoft and GitHub **stakeholders** in Agile sprints from requirement analysis through deployment and ongoing production support for the platform.
- Wrote 20+ unit and integration **test** suites with PyTest in nightly Jenkins CI/CD, and led an AWS Glue and S3 ETL redesign that boosted pipeline efficiency 68% and cut storage costs 15%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Translated client requirements into an end-to-end architecture with 12+ integration points, **presented** it to the client for sign-off, and delivered Python Flask booking and payment APIs at sub-200ms latency.
- Integrated PCI-compliant multi-currency **payments** with fraud checks and idempotent webhooks for refunds and disputes, cutting manual payment handling by 70% and powering hourly Databricks analytics for internal dashboards and client reporting.

Skills

Languages: Python, TypeScript, JavaScript, SQL, Java, C++
AI Engineering: LLM Orchestration, RAG, Embeddings, Vector Retrieval, FAISS, Pinecone, Prompt Chaining, Model Evaluation, Fine-Tuning, Claude, GPT-4o
ML: PyTorch, Transformers, Reinforcement Learning, Weights & Biases, Hydra
Engineering: FastAPI, Flask, Node.js, React, REST APIs, PostgreSQL, Git, GitHub Actions, CI/CD, PyTest
Cloud: AWS, Azure, GCP, Docker, Databricks
Delivery: Claude Code, AI-Assisted Development, Client Presentations, Requirements Gathering, Technical Writing

Projects

Insurance Claims Triage Agent with Evals | *Python, Anthropic SDK, MCP, LangGraph, pgvector, LangSmith*

- Designed a LangGraph **agent** that reads claim documents, retrieves policy clauses with **RAG**, calls claims-system tools over MCP, routes edge cases to adjusters, and gates releases on a LangSmith eval set.

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, Embeddings, PostgreSQL*

- Built a deployed **RAG** platform with transformer embeddings and vector retrieval over FAISS and Pinecone across 50K+ documents, reaching 95% query accuracy and cutting document review time 60%.

CaptionCraft Live | *PyTorch, Transformers, BLIP, GIT, ViT-GPT-2*

- **Fine-tuned** and benchmarked BLIP, GIT, and ViT-GPT-2 against a custom model on 10K images and 100K captions, raising caption accuracy from 81% to 96% behind real-time inference APIs.